

1. Flowchart / System Architecture Notes

Conceptual Notes:

- Professional systems are designed before coding begins.
- Flowcharts help organize the workflow of the system.
- Validation steps ensure data quality and system reliability.
- Error-handling prevents the system from crashing when APIs or data fail.
- Swimlane architecture separates system responsibilities into layers:
 - Data Engineering
 - NLP / ML Services
 - Analytics / BI

Simple Conceptual Code:

```
if api_connection == True:
    collect_data()
else:
    log_error()
```

```
if sentiment_score > 0.75:
    send_alert()
```

2. Twitter/X API Research Notes

Conceptual Notes:

- APIs allow systems to collect external data automatically.
- Twitter/X API requires authentication keys and tokens.
- Tweepy is a Python library for Twitter API integration.
- API rate limits control how many requests can be made.
- Tweets contain unstructured text data.

Simple Conceptual Code:

```
import tweepy

client = tweepy.Client(bearer_token="YOUR_TOKEN")

tweets = client.search_recent_tweets(
    query="TSLA stock",
    max_results=10
)
```

```
print(tweets)
```

3. NLP / Preprocessing Notes

Conceptual Notes:

- NLP stands for Natural Language Processing.
- Text preprocessing improves sentiment analysis accuracy.
- Tokenization splits text into words/tokens.
- Stopwords are common words like:
 - the
 - and
 - is
- Ticker extraction identifies stock symbols such as:
 - \$TSLA
 - \$AAPL

Simple Conceptual Code:

```
import re
```

```
tweet = "TSLA stock is going up! 🚀"
```

```
# lowercase text
```

```
tweet = tweet.lower()
```

```
# remove symbols
```

```
tweet = re.sub(r'^a-zA-Z0-9\\s$', "", tweet)
```

```
print(tweet)
```

Ticker Extraction:

```
import re
```

```
text = "Watching $TSLA and $NVDA today"
```

```
tickers = re.findall(r"\\$[A-Za-z]+'", text)
```

```
print(tickers)
```

4. GitHub / Project Organization Notes

Conceptual Notes:

- GitHub stores project code and documentation.
- README files explain the project structure and setup.
- Folder organization improves maintainability.
- Modular systems separate components into folders.

Example Folder Structure:

```
project/
|
|— data/
|— scripts/
|— dashboard/
|— docs/
|— models/
|— README.md
```

Basic Git Commands:

```
git init
git add .
git commit -m "Initial project setup"
git push
```

5. Sentiment Analysis / Dashboard Notes

Conceptual Notes:

- Sentiment analysis classifies text as:
 - positive
 - negative
 - neutral
- FinBERT is designed for financial sentiment analysis.
- Dashboards visualize analytics and trends in real time.
- Streamlit is a Python dashboard framework.

Simple Sentiment Code:

```
from textblob import TextBlob

text = "Tesla stock performance looks strong."

analysis = TextBlob(text)

print(analysis.sentiment.polarity)
```

Simple Dashboard Example:

```
import streamlit as st

st.title("Stock Sentiment Dashboard")

st.metric(
    label="TSLA Sentiment",
    value="0.82"
)
```